

54-57-2022, 16/9/22

University of Information Technology & Sciences (UITS)

Faculty of Science and Engineering

Department of Computer Science and Engineering

Program: B.Sc. in CSE

Term Final Examination, Autumn 2025

Course Title: Database Management System

Course Code: CSE315

Marks: 50

Time: 3(three) hours

(Answer all questions)

- | Q No. | | Marks |
|-------|--|-------------------------|
| 1. | <p>a) To maintain the integrity of the data, there are four properties described in the database management system, which are known as the ACID properties. Explain the <u>ACID</u> properties of a transaction.</p> <p>b) Draw an ER diagram of UITS UCAM. The diagram should contain strong entity, weak entity, key attribute, multi-valued attribute, composite attribute, derived attribute as well as cardinality.</p> | <p>[04]</p> <p>[06]</p> |
| 2. | <p>a) Consider the relational database, where the primary keys are underlined.
employee (<u>person name</u>, street, city)
works (<u>person name</u>, company name, salary)
company (<u>company name</u>, city)
Write Structured Query Language (SQL) commands for each of the following statement:</p> <ul style="list-style-type: none">i. Find employees who live in the 'Dhaka' city.ii. Display all employees who earn more than 50,000tkiii. Find the highest paid employee in 'Steel Company'iv. Find the highest paid employee in each company.v. Find employees who work for companies located in "Dhaka"vi. Count the number of employees working in each companyvii. List employees who earn more than the average salary of their company. <p>b) Explain key difference among DELETE, DROP and TRUNCATE</p> | <p>[07]</p> <p>[03]</p> |
| 3. | <p>a) A join clause is used to combine rows from two or more tables, based on a related column between them. DBMS supports many kinds of different joins including inner join, self-join, cross join, and outer join.</p> | <p>[04]</p> |

Perform left outer join, right outer join, full outer join and inner join on the Loan and Borrower table.

Loan No	Branch	Amount
L1	Dhaka	1000
L2	Barisal	2000
L3	Rajshahi	1500

Tabl-01: Loan

CName	Loan No
Alex	L1
Jack	L2
Rayan	L5

Table-02: Borrower

- b) **Apply** normalization of the following table up to the second normal form (2 NF) to minimize redundancy.

[06]

S_id	S_name	Age	Course	Post Code	City
2501	Khan	21	CSE 329, CSE 330	4558	Rajshahi
2502	Choudhury	22	CSE 329	6550	Pabna
2506	Islam	23	CSE 329, CSE 330, CSE 327	3200	Barishal

Table-03: Student Information

4. a) An attribute is a property or characteristic of an entity. An entity may contain any number of attributes. An entity may have different types of attributes.

[06]

Explain five types of attributes and explain each with suitable example with figure.

- b) Mapping cardinalities defines the relationship between numbers of entities in one entity set with the number of entities to other entity sets

[04]

Give an example of each of the four mapping cardinalities.

5. a) Most Redundant Array of Independent Disks (RAID) levels employ striping, it is a term used to describe when individual files are split and written to more than one disk.

[06]

Describe characteristics of each RAID level.

- b) When accessing the database, anomalous situation may occur where the database may not be in expected state.

[04]

Describe different types of Anomaly in database management system with an example.